

White Paper

Where to Deploy Generative AI Solutions in Life Sciences

*Learning from common pitfalls in
commercial generative AI efforts*

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Introduction

Generative AI (GenAI) has emerged as a transformative technology, promising unprecedented capabilities across industries. For commercial teams in life sciences, the potential applications are vast, ranging from faster insights to drive investment decisions to increasing automation for labor-intensive manual tasks, such as marketing content creation. With the speed of GenAI technology development and an abundance of bold claims, business leaders can struggle to separate use cases with big impact from fanciful concepts of limited value.

Too many pilots and proofs-of-concept fail to work as intended. **Across industries, more than 70% of GenAI experiments fail to make it to production.**¹ The expanding landscape is quickly moving on from prompt-engineering to a multitude of differentiated capabilities and integrations with other technologies.

General discourse and think-pieces on the shortcomings and risks of GenAI cover many of the same now-familiar themes: hallucinations, data leakage, privacy concerns, etc. While there is clearly

some validity in these challenges, these are symptoms of technological infancy that are likely to resolve themselves as GenAI matures, and they do not speak to the essential underlying problems. This white paper will address some of the core reasons why GenAI applications for life sciences commercial teams are stumbling, offering guidance and practical rules of thumb to prioritize the best use cases and apply best practices.



UNDERSTANDING GENERATIVE AI

GenAI uses Large Language Models (LLMs) that are trained on vast datasets to recognize patterns and generate human-like text, including computer code. LLMs have the ability to understand context and content to produce coherent, relevant text in response to text input (prompts). Most LLMs come pre-trained on vast amounts of general knowledge that gives emergent capabilities, such as the ability to solve certain reasoning problems or translate between languages.

Despite the name, LLMs extend into non-text domains. Multi-modal LLMs create audio, images or short videos that are transforming content creation. LLMs also exist for technical or scientific tasks, such as predicting the results of chemical reactions, though few examples of this sort are currently suitable for commercial purposes.

Combining LLMs with other tools, such as search engines or statistics engines, further extends their abilities. Like many companies in 2024, IBM issued a “reality-check” for more realistic expectations of LLMs while predicting the continued growth of agents and automation.² Agentic AI, which is empowered to perform dedicated tasks autonomously, is expected to dramatically augment what GenAI is capable of.³

Core reasons why GenAI applications sometimes fail

At IQVIA we have seen a broad range of GenAI programs ranging from our own early investigations to programs designed by our clients. Some have found instant success while others have struggled. From these internal and external observations, our experts identified four major reasons that programs can fail and use these to identify potential problems early and guide our own GenAI product development.

1. Unrealistic expectations that GenAI can create deep insights from limited data

- » Most business use cases ask the models to process data or information source(s) about your business. Implementation teams add guardrails (rightly so) to ensure that responses relate to the source information and avoid hallucinations. LLMs are unable to create insights if the information necessary for those insights isn't somewhere in the data sources.

2. Implementations don't fit within users' workflow

- » Inconvenience kills value. Implementations need to insert GenAI use cases into existing workflows to maximize adoption and thereby business impact. Forcing users to switch between software applications and endure slow response times can hammer adoption rates.
- » A positive user experience matters. Marketing research has shown that online shoppers are apt to abandon purchases that involve too many clicks or unresponsive pages. While users will tolerate more friction for essential work tasks, it's human nature to resist processes that feel bothersome. All GenAI implementations should streamline user experience to the degree possible.

3. Challenges to prove required accuracy

- » While computer programs typically are deterministic and create predictable error types (e.g. return no data or crash), errors in GenAI are less predictable and harder to understand when validating a new service. Measurement accuracy itself can be challenging and subjective for GenAI.⁴

4. Not all use cases are useful for end users

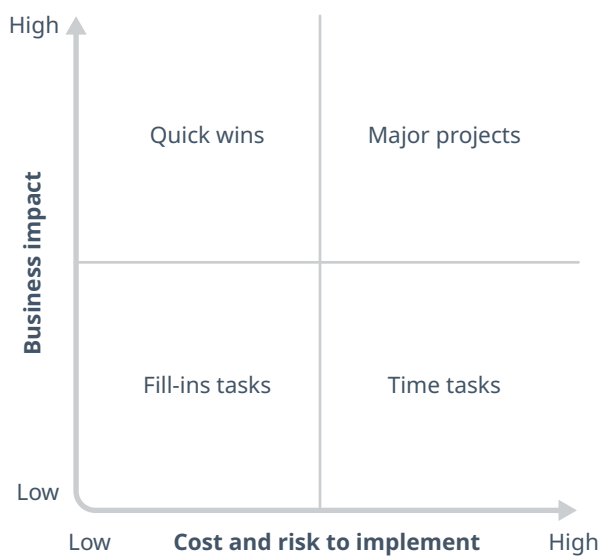
- » In the excitement to find places to deploy generative AI, some use cases have failed to enthuse end users. This is especially true for expert users who have become fast and effective at using systems, creating a high barrier to delivering improvements.

Prioritization framework

We use the common prioritization matrix that rates each use case or project for impact and effort. Business impact is ideally the expected value of revenue generated and costs saved, though it could be customer satisfaction, user adoption or other secondary measures. Because GenAI is relatively new, project failure rates are higher than other tech projects, so we include Risk as a factor on the x-axis along with the Cost.

Multi-million-dollar investment decisions warrant detailed financial modelling for risk-adjusted investment, revenues and savings. Benchmarks and back-of-the-envelope calculations can create a simpler business case to help management make informed decisions. With a lack of benchmarks for GenAI programs, or as a first filter for ideas raised in a brainstorming session, we provide heuristics and rules of thumb to help you identify the most promising GenAI opportunities.

Figure 1: Prioritization matrix



Business impact

How to spot GenAI use cases with high business value



Repetitive manual tasks

- LLMs can rarely do things a person can't do if given enough time. The vast majority of successful GenAI products automate human-like tasks vs. doing something completely novel
- In a cross-industry survey, Deloitte found that >50% of executives identified cost, efficiency or speed as the single biggest benefit of GenAI⁵



Large volumes of text or code

- LLMs excel at text analysis with transformative capabilities when compared to older Natural Language Processing (NLP) or statistical techniques
- Compared to text-based use cases, GenAI for audio and visual data is less common in life sciences commercial organizations. Audio and visual uses are largely limited to marketing content and employee training, although this may change
- GenAI performs especially well to accelerate coding tasks, including for advanced analytics. In October 2024, Google announced that more than 25% of its new code is being generated by AI and reviewed by engineers.⁶ This trend is certain to continue as the value of AI-assisted programming becomes undeniable



Intrinsic business value

- A process that has a bigger impact on the P&L will have more business value. For example, a tool to evaluate companies for M&A may get less frequent use but have more value than an AI tool that generates new PowerPoint templates

TIP: HOW TO IDENTIFY THE MOST REPETITIVE MANUAL TASKS TO AUTOMATE

A task may be frequent and unique to your business, requiring a bespoke implementation. One example may be a tech-enabled sales process custom-built to give your company a sales edge. Another is responding to inquiries and support requests where the responses need fine-tuning to answer accurately in the context of your company's systems and business.

Other tasks may not repeat often for your company but repeat often across your industry. An individual pharmaceuticals company may have only a handful of drug candidates going through regulatory filings, but across the industry there are thousands. Specialist companies such as IQVIA can use that data volume to develop GenAI products to automate parts of the regulatory filing.





Cost

How to spot GenAI use cases with lower or higher cost



Data and product engineering

- Degree of change for user interfaces and/or technical workflows
- Number and complexity of data sources to incorporate or analyze



Change management

- Cost scales with the number of users, and the amount of training, comms and leadership support needed to drive successful change
- A great user experience substantially reduces effort for change management



Compute

- LLM compute costs can escalate with number of users, query token length, and choice of model
- More capable smaller LLMs have reduced costs to the extent that they affect the business case for only the largest-scale programs with high token counts



Inclusion of Personally Identifiable Information (PII)

- Any PII requires additional work with privacy, legal and IT to ensure data remains protected, adding to implementation cost and timelines

Risk

How to spot GenAI use cases with lower risk of failure



Output reviewed by a human

- People do not yet trust GenAI to completely automate important tasks. Output needs a human sense check before it's used for a decision or business process



High-quality user experience

- Is there an obvious place to insert the GenAI service into an existing platform or workflow (e.g. email) vs. deploying a standalone tool?



Users are not experts

- Expert users have spent years or decades learning how to use their preferred tools quickly
- Expert users have a low tolerance for the limitations of any new tools

Future developments for GenAI technology

In the next few years across all industries, GenAI technology is poised for remarkable advancements that will expand its capabilities. ChatGPT o3 and o3-mini give a taste of more sophisticated reasoning abilities that will enable AI systems to perform complex analytical tasks. Open weights models, especially Llama models from Meta, are democratizing LLM developments to allow a proliferation of LLMs fine-tuned for different purposes.

Small Language Models (SLMs), such as the Phi models from Microsoft and Qwen models from Alibaba, can deliver “good enough” quality for some improvements in speed and substantial reduction in running cost.

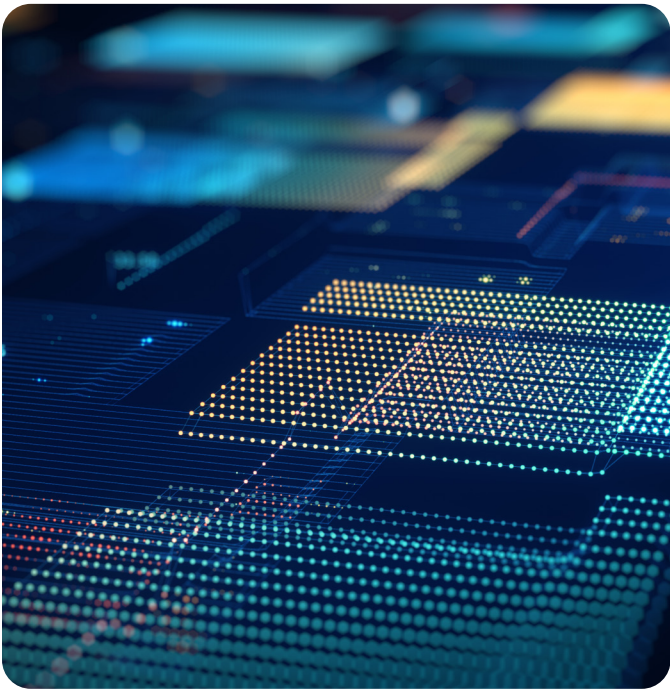
New technologies typically evolve through a set of three phases shown in Figure 2.

Figure 2: Three phases of technology optimization

 CUSTOM	 MODULAR	 INTEGRATED
By necessity, different companies create their own solutions where no industry standard options exist. <i>Think of '80s era personal computers with custom parts specific to each manufacturer.</i>	Driven by efficiency and companies specializing in narrow parts of the value chain, parts become interchangeable with industry-defined standards. <i>Think of '90s era desktop computers with user-replaceable options for CPU, hard drive, etc.</i>	For more efficiency and convenience, parts converge into packaged offerings, reducing complexity for the end user. <i>Think of a modern Apple Mac laptop that is more convenient and powerful but not upgradeable.</i>

We will see more integration of AI into existing products and services for more seamless workflows, minimizing the learning curve for less-expert users. Development frameworks such as LangChain, Llama Index and Auto-GPT are streamlining developers’ ability to integrate GenAI into applications. As the track record of GenAI builds and improves, companies will identify lower risk use cases to leverage AI with only automated quality controls and human spot-checks.

While some companies such as Microsoft have been able to integrate GenAI quickly into their software products, most are still in the custom phase. Tailored solutions for business processes will often need custom or modular solutions, and industry standards will emerge to drive inter-operability, efficiency and compliance.



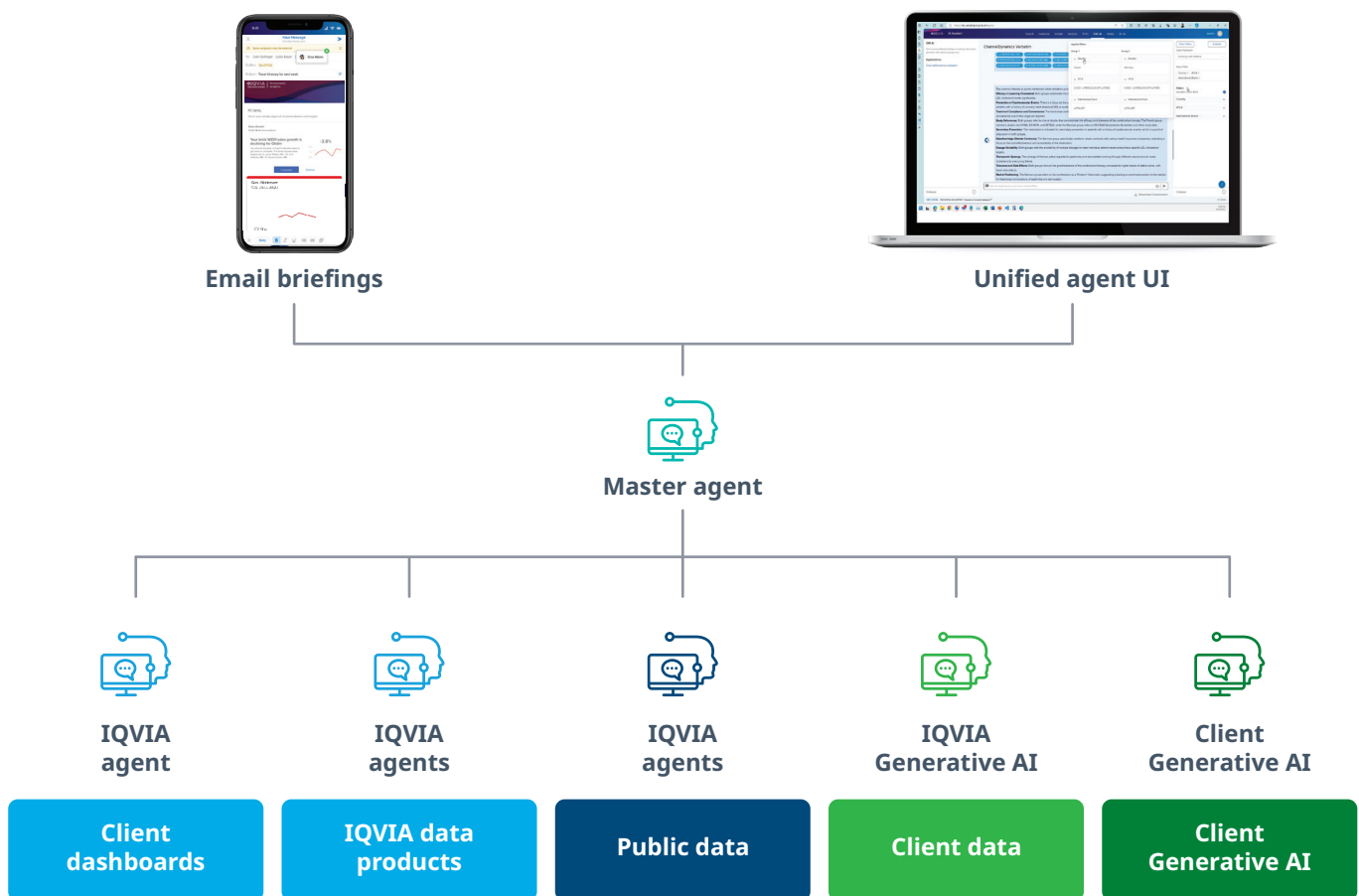
IQVIA AI ASSISTANT

IQVIA's ecosystem of GenAI agents share a common *Healthcare-grade AI™* technology stack and user interface. This approach allows each agent to specialize for specific tasks, giving higher accuracy, better performance and shorter development timelines. An agent orchestrator, or master agent parses plain text requests from the end user, sends requests to the appropriate agent(s), and synthesizes the returning responses into a single output to the user.

IQVIA licenses these tools to clients SaaS and on-premise software products, and we use them internally to enhance and accelerate project delivery. Many of these agents are also embedded in syndicated data products, optimized for those data sets and available for instant deployment to thousands of users.

To future-proof the approach and deliver with more flexibility, agents include API interfaces to call each other and interact with our clients' GenAI solutions. IQVIA's master agent will collate and summarize responses from other agents, including client agents exposing an API. Or our clients may have their own agent call the APIs on IQVIA's licensed agents to perform a similar task.

Figure 3: IQVIA's agentic approach to GenAI product development





Conclusion

Generative AI holds immense promise for commercial teams in the life sciences industry. By following practical rules of thumb to identify the best mix of business impact, cost and risk, organizations can identify and implement the most impactful use cases. Through successful integration of GenAI into important business processes, life sciences companies can drive innovation, enhance patient outcomes, and achieve operational excellence.

To explore how IQVIA's Healthcare-grade AI™ can transform your business, contact us at iqvia.com/oa

About the authors



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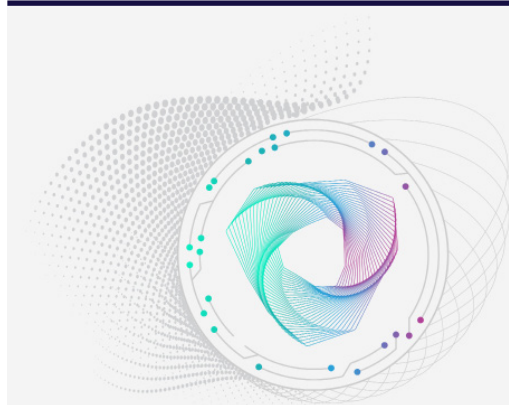
Charles Rink has over 20 years of experience in strategy consulting, technology, and analytics for the life sciences industry globally. Based in London, he brings extensive experience advising clients in commercial strategy, analytics automation, and omnichannel operations globally. He holds a BS degree in Molecular Biochemistry & Biophysics and Economics from Yale University.



WILLIAM O'REILLY

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Will leads the Generative AI and Machine Learning product capabilities across Orchestrated Analytics, including the IQVIA AI Assistant. Hailing from a diverse background as a clinical doctor, healthcare data analysis consultant, and data product manager, Will leverages his varied expertise and deep curiosity to bring a unique perspective and create innovative solutions in Life Science technology products and services.



About IQVIA *Healthcare-grade AI*[™]

IQVIA *Healthcare-grade AI*[™] is AI engineered to meet the level of precision, speed, and trust required in life sciences and healthcare. Healthcare-grade AI is made possible with IQVIA Connected Intelligence[™], which connects unparalleled quality healthcare data, analytics, technology, and deep expertise to train, optimize, and harness the full potential of AI.

Drive new levels of performance, enhance efficiencies, and improve patient outcomes all over the world. [Learn more at IQVIA.com/AI](https://www.iqvia.com/AI).

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